The 6 biggest eBPF beneficiaries

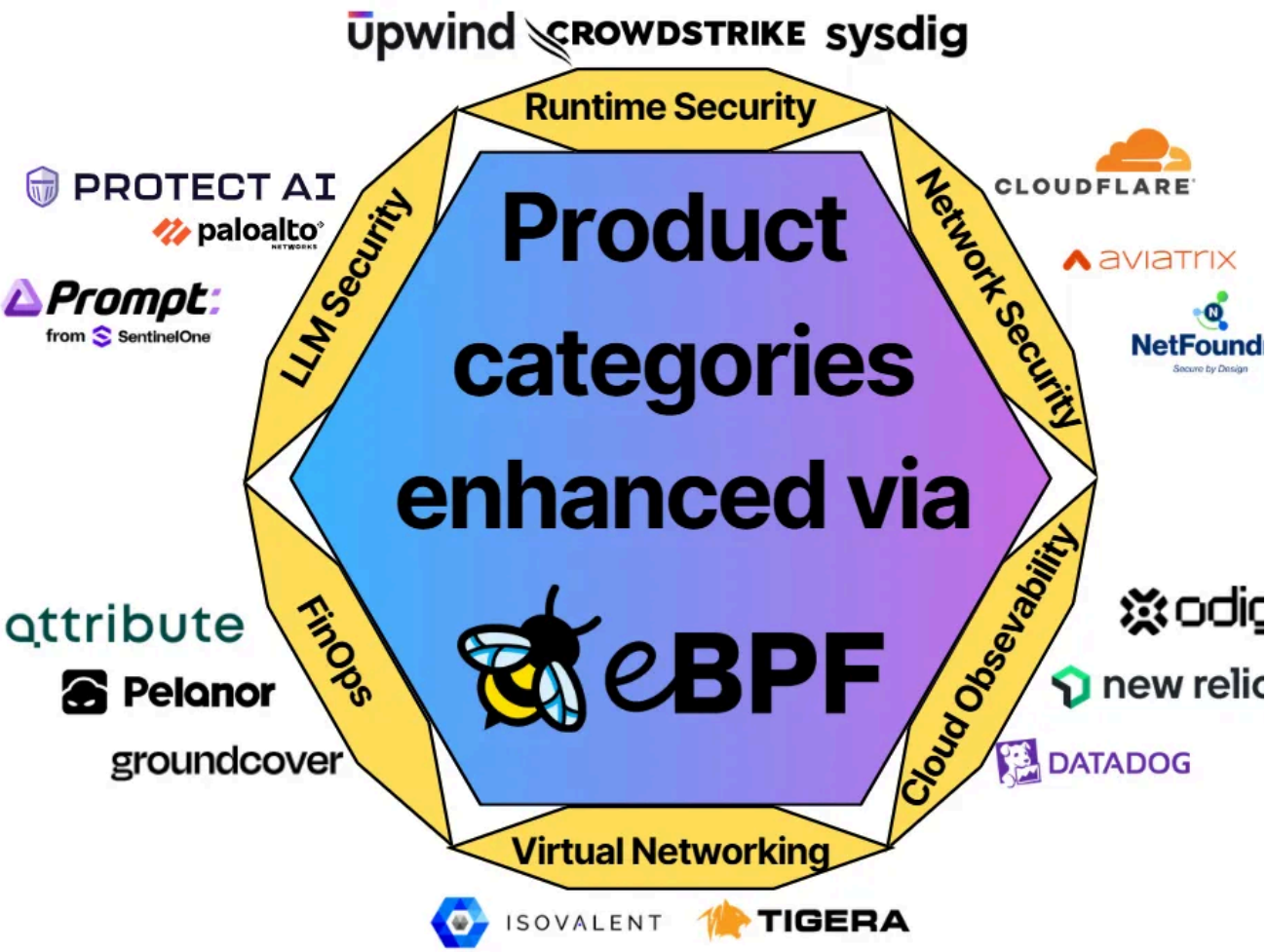
ANDREW GREEN
FEB 09, 2026

2

I've spent a few years looking at how products use eBPF, but only when I wrote t eBPF for the Infrastructure Platform paper for the eBPF foundation, I saw exactly used eBPF the most and/or the best.

So here are the top 6 categories that benefited most from using eBPF

- The most considerable advancements - Virtualized Networking
- The highest adopters - Runtime Security
- The best performance improvement - Network Security
- The expected ones - Cloud Observability
- The unexpected ones - LLM security
- The cooles - FinOps



The most considerable advancements - Virtualized Networking

There are many flavors of virtualized networking, including software-defined networking for on-premises networking, cloud networking for orchestrating hyperscaler networks, and container networking for Kubernetes.

Virtualization introduced enormous flexibility into networking, but also an invisibl overhead with each layer of abstraction. By letting packet processing, policy enforcement, and routing logic happen directly in the kernel, **eBPF can improve performance and scalability of virtualized networks**, realizing performance g at the software level that are otherwise unachievable with traditional cloud and container networking constructs.

Some of the best implementations of eBPF are done at he CNI level:

- Cilium is an open-source container networking interface (CNI) that is synony with eBPF and has been designed on top of it to address the networking, se and visibility of containers.
- Project Calico is an open-source CNI that offers a pluggable eBPF-based da plane for Kuberentes networking and security.

The best performance improvement - Network Security

Instead of routing packets through external appliances, **packet inspection and filtering are done at the kernel level on each endpoint**, and are attached to ne interfaces at both ingress and egress points using the Linux traffic control subsy This creates distributed firewalls and enforces microsegmentation. eBPF-based network security offers a wide range of technical features, including Layer 4/7 st filtering, reputation-based detection, microsegmentation, IP masquerading, and

This shift means network security no longer needs to live in a box, physical or vir With eBPF, it becomes **an integral, programmable layer of the operating syst itself**, enabling zero-trust microsegmentation that scales as seamlessly as the workloads it protects.

Some examples include the following:

- Aviatrix implements eBPF into the Distributed Cloud Firewall product to anal observe and enforce network security policies for traffic within and between environments.
- Cloudflare's eBPF-based DDoS protection that blocked a 7.3 Tbps attack
- Netfoundry's ZittiFirewall is a standalone firewall application utilizes both tc- and xdp to provide stateful IPv4/IPv6 firewalling.

The highest adopters - Runtime Security

Several types of solutions secure processes or applications as they are executin most notably being endpoint detection and response (EDR) and the proposed ne generation extended detection and response (XDR).

As EDR was only designed to protect endpoints, a more comprehensive solution attempted to bring runtime security to clouds, networks, and virtualized applianc However, the agents used to enforce security policies consume a high percentag resources on the hosts they run on and require continuous management. The solutions that do not use kernel modules have their visibility and enforcement restricted by the kernel data made available via APIs.

The **runtime security landscape has quickly adopted eBPF** and evolved into r robust products, where **enforcement and visibility are achieved by introducir dynamic, secure, low-overhead, and efficient programs that run directly in kernel**. Some examples of categories that heavily use eBPF for runtime security cloud detection and response (CDR) and application detection and response (AI

- Crowdstrike's Falcon Data Protection for Cloud: Delivers runtime protection cloud data at rest and in motion, leveraging eBPF to detect and block unauthorized data movements in real time, without slowing systems down.
- Sysdig's Falco is an open source, cloud native runtime security tool that des to detect and alert on abnormal behavior and potential security threats in rea time.
- Upwind's eBPF sensor creates prevention policies automatically stops malic processes from running in workloads

The expected ones - Cloud Observability

Observability has long depended on agents, sidecars, and SDKs with each one sampling metrics, logs, and traces from different parts of the stack. But stitching these perspectives together has always meant gaps, latency, and overhead. **eBF collapses that complexity by operating from a single vantage point in the ke** Some of this richer kernel-level data includes distributed traces, service mappin

protocol-specific usage such as HTTP, gRPC calls, SQL queries, Kafka, system c network interface events, and performance data **all without modifying a single of code**.

For organizations with an on-premises footprint, eBPF also provides hardware-le observability, such as visibility over memory access patterns, disk I/O performan and network interface behavior. This kernel-level visibility helps identify hardwar contention, resource misallocation, and performance degradation across physica servers, storage devices, and network adapters. This is particularly valuable for optimizing on-premises data centers where operators must understand the relationship between software behavior and underlying physical hardware.

For hybrid environments, this model is particularly powerful. Whether workloads on-premises or across multiple clouds, **eBPF delivers consistent telemetry frc same kernel hooks, eliminating the blind spots** that arise from mixing vendor agents and hypervisor-based monitoring. The result is a unified, event-driven vie the entire system.

- Datadog used eBPF within build Universal Service Monitoring (USM), where eBPF agent parses HTTP(S) traffic from the kernel network stack.
- New Relic acquired Pixie, an open source Kubernetes-native-in-cluster observability platform that provides instant visibility into Kubernetes workloa
- Odigos uses eBPF to automatically generate traces in OpenTelemetry forma Their performance tests claim that eBPF-based automatic instrumentation is 20x faster than manually instrumenting code.

The coolest - FinOps

To calculate cloud costs without relying only on data exposed by hyperscalers, e probes are used to monitor system calls, payload level network traffic, and applic behavior in real-time. eBPF can help determine how cloud services are used and this back to kernel events.

By observing CPU scheduling, network paths, and I/O patterns in real time, **eBPI makes invisible inefficiencies visible**. This lower-level data can also differentia between tenants on shared services or multi tenant environments. It can also me that costs can be attributed to each cloud service such as managed databases, storage, or egress costs.

- Pelanor's eBPF Kubernetes sensor delivers granular cost allocation for every cluster, namespace, workload, and even individual network endpoint.
- Attribute correlates the granular data collected by eBPF probes with busines metrics to calculate costs and provide actionable insights over cloud spendi
- Groundcove uses eBPF to track resource usage across all K8s clusters for resource consumption monitoring to track spending in real time.

The unpredictable ones - LLM Security

Protecting GenAI applications, specifically those that use LLMs, is a new challen security architects. Most tools designed to address LLM security issues have **di opted for eBPF to trace the interactions** between applications and external lar language models.

By instrumenting eBPF at the network layer, these solutions monitor and log the calls made to these services, capturing information such as request and respons data, latency, and error rates. Some examples include:

- Prompt Security (Acquired by SeninelOne) uses eBPF to trace the communi between applications and the vector database, providing insights into query patterns, response times, and potential issues.
- Protect AI (acquired by Palo Alto Networks) uses eBPF to discover AI applica usage.

Thanks for reading Green Abstracts! Subscribe for free to receive new posts and support my work.